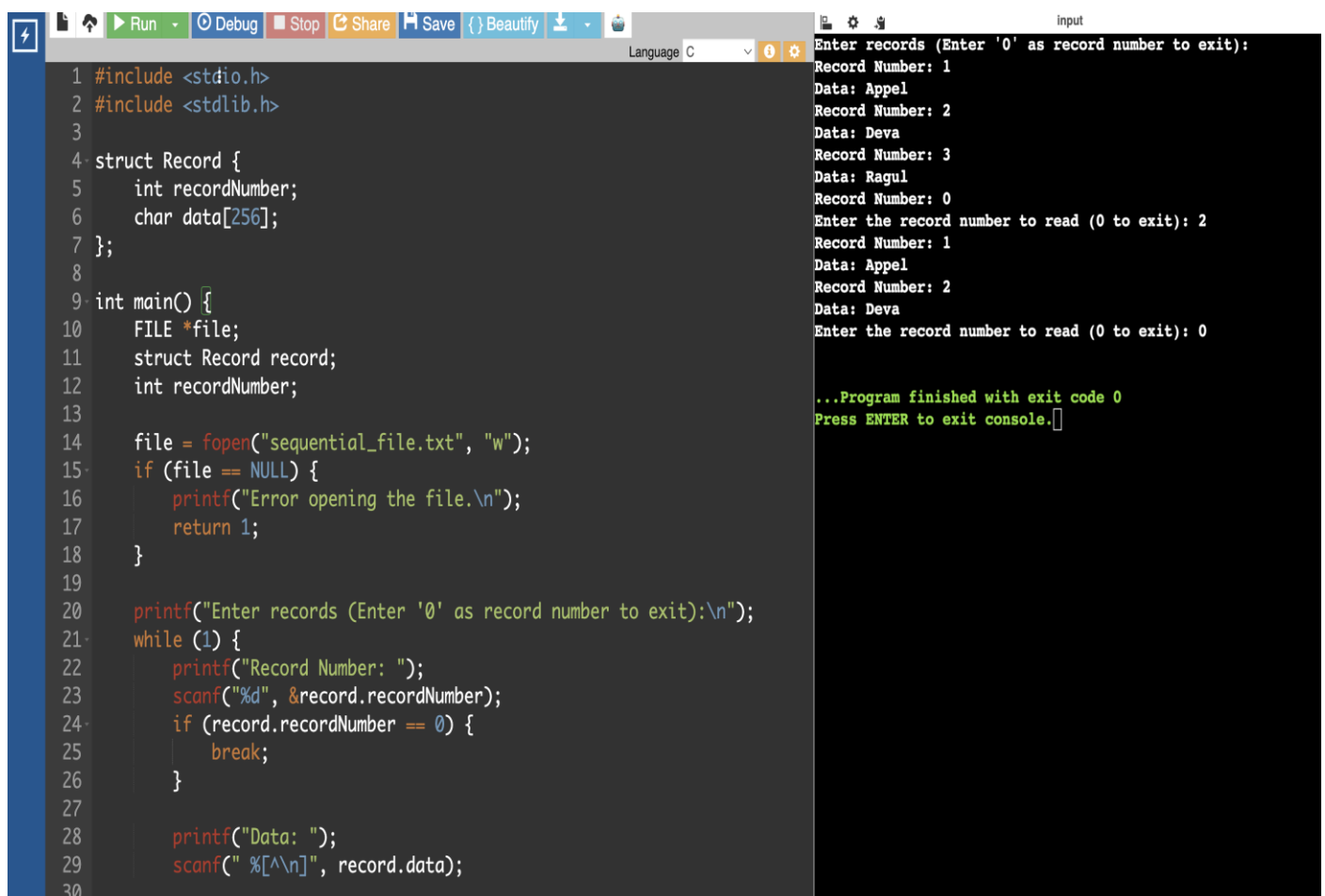


EXPERIMENT 35:

QUESTION

Consider a file system that brings all the file pointers together into an index block. The entry in the index block points to the i th block of the file. Design a C program to simulate the file allocation strategy.

PROGRAM:



```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Record {
5     int recordNumber;
6     char data[256];
7 };
8
9 int main() {
10     FILE *file;
11     struct Record record;
12     int recordNumber;
13
14     file = fopen("sequential_file.txt", "w");
15     if (file == NULL) {
16         printf("Error opening the file.\n");
17         return 1;
18     }
19
20     printf("Enter records (Enter '0' as record number to exit):\n");
21     while (1) {
22         printf("Record Number: ");
23         scanf("%d", &record.recordNumber);
24         if (record.recordNumber == 0) {
25             break;
26         }
27
28         printf("Data: ");
29         scanf("%[^\n]", record.data);
30
31         fprintf(file, "Record Number: %d\nData: %s\n", record.recordNumber, record.data);
32     }
33     fclose(file);
34     printf("...Program finished with exit code 0\n");
35     printf("Press ENTER to exit console.\n");
36 }
```

Enter records (Enter '0' as record number to exit):
Record Number: 1
Data: Appel
Record Number: 2
Data: Deva
Record Number: 3
Data: Ragul
Record Number: 0
Enter the record number to read (0 to exit): 2
Record Number: 1
Data: Appel
Record Number: 2
Data: Deva
Enter the record number to read (0 to exit): 0

...Program finished with exit code 0
Press ENTER to exit console.